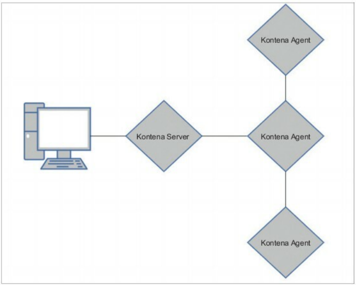

Figure 1: The Kontena system

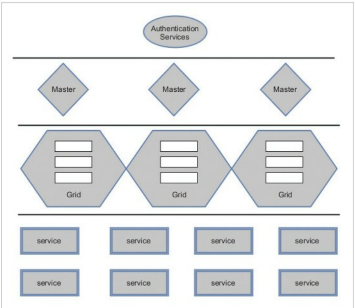

Figure 2: Kontena architecture

Features of Kontena

Listed below are the key additional features of Kontena, apart from its standard features:

- Separate authentication layer
- Audit logs
- Built-in Docker image registry
- Remote access to workload services
- Load-balancing
- Log and statistics aggregation with streaming
- Different roles and access control for users
- A scheduler with affinity filtering

Installation and a demo

Kontena is built using Ruby, so the latter first needs to be installed. We can download the latest version from the official site. What we require is version > 2.1 (*https://www.ruby-lang.org/en/documentation/installation/*).

If you are on Windows, then use the link *https://rubyinstaller.org/* to download the Ruby installer.

The next step is to install Kontena-CLI. We can download it using the Rubygems Package Manager, as follows:

```
gem install kontena-cli
```

To check the Kontena version, run the command given below:

```
kontena Version
```

To install Kontena Masters, we can either use the cloud platform or the local environment. Here, we will consider the local environment.

We will use Vagrant to install Kontena Master in a local environment. Here, too, the minimum requirement is version 1.6 and above.

The two links given below provide the details for installing and configuring Vagrant, so we will not go too deep into the topic.

- *https://www.vagrantup.com/*
- *https://www.vagrantup.com/docs/installation/index.html*

Once you are ready with Vagrant, we will start installing Kontena Master, using the following commands:

```
kontena plugin install vagrant
kontena vagrant master create
```

While installing, you will be asked how you want to authenticate it. It's good to choose *External authentication*. We can also use Kontena Cloud for the authentication.

The next step is to install Kontena nodes; we will use Vagrant to do this:

```
kontena grid create newgrid
# or to switch to an existing grid, use:
kontena grid use newgrid
```

Now the grid is ready, and we have selected *newgrid* as the current grid. We will now create nodes in it:

```
kontena vagrant node create
```

To create multiple nodes for faster computing, we can repeat this step. It is recommended to have more than one node in the grid.

To check all the nodes in the grid, we can use the command given below:

```
kontena node list
```

Now we are ready with the environment, and it's time to deploy our first application stack.